Breaking down the basic stats of characters in "CS2"

In this article, we break down the basic values of characters in CS2, including character base size, movement ability in each state, and character hit range.

In the disassembly method, we mainly use the CS2 console to obtain real-time location information and speed information from the player using command cl_showpos 1, and perform approximate calculations to obtain a series of character values.

The numerical units in the full text are unified as the custom unit "unit" in CS2, and the definition is abbreviated as "u ".

1.1 Character size value

First, the character's basic body shape value is calculated. Thanks to the ability of players to stand on top of each other in CS2, we placed a static bot in the scene that was on the same level as the player, and calculated the Z-axis position of the player standing on the bot and the horizontal plane The difference is the height value of the character.

Stand on a horizontal surface	
Stand on top of the crouching bot	
Stand on top of an upright bot	

Calculating the difference, we can get the following values:

Character standing height	72.03u
Character crouching height	54.03u

Next, get the character width. From the data, it can be seen that the character collision volume in CS2 is a square box, so we can still borrow the position difference for approximate calculations:

Stand to the left of the bot	

Stand to the right of the bot	
Stand in front of the bot	
Stand behind the bot	

The difference is still calculated to obtain the following values:

	Double the value	haploid values
Character width left and right	64.09u	Approx.32u
Character front and back width	64.13u	Approx.32u

It can be seen that in CS2, the left and right widths of the characters are equal to the front and rear widths. To sum up, we get the following character size values, approximate to single digits:

Character standing height	72u
Character crouching height	54u
Character width	32u

1.2 Character collision volume

Controlling the character to perform a close crash test with bots on both sides , we can find:

- The character and other players on both sides have collision volumes and cannot blur through;
- The collision volume is approximately cube, and there is obvious angular perception during the collision process.

1.3 Character movement ability

Next, disassemble the character's jump height under different operations and the difference in the character's movement speed when holding different items to obtain the character's basic movement attribute value.

In the console command cl_showpos , we can directly get the speed and jump height attributes of the knife character, so we only need to test them in a row.

1.3.1 Character jumping ability

First, test the character's jumping ability, we test the player's different heights in jumping, squatting, and jumping squats:

Jump directly	jump: 0.000 (55.83)	Press the space separately
crouch jump	jump: 0.000 (66.00)	Before fully crouching Press the space to jump

Fully crouch jump		After crouching completely Press the space
Jump first and then squat		Hold down crouch after jumping
Squat first and then jump		After crouching completely Release and press the space

Through the multi-condition test, we can find that crouching has a large and complex effect on the jump height of the character:

- Squat jumping operations can make players jump higher;
- Pressing the squat button during the jump will affect the jump height;
- Improper crouching maneuvers have a negative impact on jump height;
- The stance and mechanics for crouching and full crouching also differ；
- Continuous jumps will cause the player's jump height to drop and fluctuate in a certain range (fatigue value).

For the purpose of statistical statistics, we take the direct jump height and the optimal squat jump height as the ideal character jumping ability:

Jump height	55.83u
Squat jump height	66.00u

1.3.2 Character movement ability test

First, the player's basic movement ability test can measure the player's maximum standing movement speed and maximum crouching movement speed:

Maximum standing movement speed	250
Maximum crouching movement speed	85

Next, we test the character's maximum movement speed when holding different items, and as an indicator of the character's movement ability, we use the fastest movement speed in the game - the movement speed of the knife as a benchmark, to calculate the movement speed penalty of different handheld objects.

Test the T faction item first:

The character holds an object	Character maximum movement speed	Speed up punishment percentage
Knife, C4	vel: 0.00 (250.01)	0%
Explosives	vel: 0.00 (245.00)	2%
Glock, dual-wielding Beretta, P250, Tec9, Bison	vel: 0.00 (240.00)	4%
MP5	vel: 0.00 (235.01)	6%
Sand Hawk, stun gun, SSG 08, UMP45	vel: 0.00 (230.00)	8%
Nova, MP7	vel: 0.00 (220.00)	12%
Galil AR, AK47	vel: 0.00 (215.00)	14%
SG553	vel: 0.00 (210.01)	16%

AWP		20%
Negoff, SG553 (open scope).		40%
APW (open scope).		60%

At the same time, the test in the crouching state was carried out, and the penalty percentage was found to be the same, so it was not repeated. We then test items exclusive to the CT faction:

The character holds an object	Character maximum movement speed	Speed up punishment percentage
USP、FN57		4%
M4A1		10%
Famas		12%
SCAR-20		14%
SCAR-20 (open scope).		52%

In summary, we can derive the character's movement speed and penalty percentage when holding different items as indicators of the character's ability to move.

At the same time, we can find the following characteristics from the test:

- The character movement speed does not directly reach the maximum/0 value, there is a process of speeding up and slowing down;
- When the character holds down a single arrow key, the mouse steering causes a slowdown, which is related to the steering amplitude, with the slowdown being most noticeable when turning 180°, which is about 50%.
- When a character holds down a single arrow key, holding down the vertical arrow key will cause a small percentage of slowdown;
- When a character holds down a single arrow key, releasing and clicking the opposite arrow key will quickly reduce the movement speed to 0 (emergency stop effect);
- After the player jumps to the ground, there is a significant movement speed penalty.

From these tests, we can conclude that CS2's movement speed is not directly numerically assigned, but rather a continuous change with a speed curve that is somewhat consistent with physical effects.

1.4 Character hit volume

Next, we test the character's basic hit volume. We use the command sv_showimpacts to show the bullet landing point, a rough test of the character's hit volume.
The following is a list of typical features:

	Decorations around the character have no hit volume

	There is no volume between the character's crotch
	The character's elbow bend space has no hit volume
	The character's feet also have a hit volume

Shooting the bot and we can find the following characteristics:

- The impact volume of the character is close to the edge of the character model, and there is no epitaxial impact volume.
- There is no hit volume between the character's crotch, no hit volume in the elbow bending gap, and the hitbox is unified with the character's limb model;
- The decorations around the character have no hit volume;

We then use the command cl_ent_hitbox to get the character's exact hit volume:

The character stands in front	

The character stands on the side	
The character crouches in front	
The character crouches on the side	

From this, we can find that the impact volume of "CS2" is composed of several capsules, which fit closely with the character model, but the accuracy is not high, and each torso part

is only composed of one capsule, so there are some blank hit areas, such as the knee joints, and some character models have no impact areas on the edges.

Figure 1 Blank area of impact at the character's knee

At the same time, different operator skins were used for testing, and it was found that the character's attack volume was consistent, ensuring fairness.